

SEMESTER PROJECT

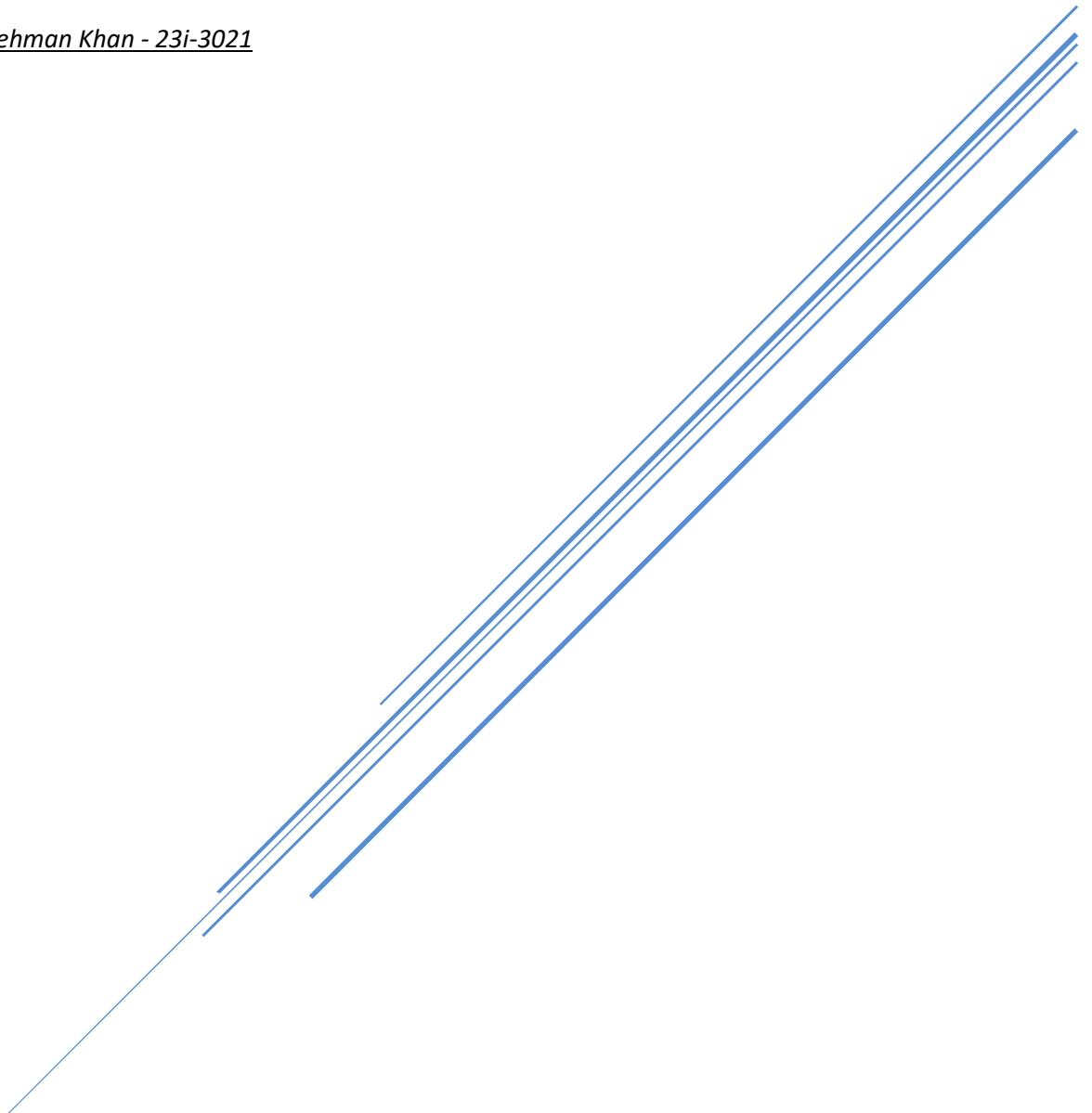
Multi-Variable Calculus

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Objective of the Problem

The main objective of this problem is to have us analyze the behavior of the demand and price of BnB with multiple factors influencing it. This not only includes the price of the product itself which may affect its demand but also the behaviors of products competing with BnB. We are to produce a multivariable function that details the demand of BnB and analyzes its behavior when the price of its competitors is the same as BnB itself and when its price is increased while the competitors stay the same. We are also to observe and report the effects of the increase in price of product A on the demand for BnB. After that, we have to suggest measures to the company that will help increase BnB's demand.

By-Hand Solution of the Problem

Task 1:

Task -1:

Computational Graph:

$$D = 10000 - 2.5 P_{bnb}^2 + 3 P_a + 1.5 P_b$$

```
graph LR; Pbnb --> q["q = 2.5 * Pbnb * Pbnb"]; Pa --> r["r = 3 * Pa"]; Pb --> s["s = 1.5 * Pb"]; q --> D["D = 10000 - q + r + s"]; r --> D; s --> D;
```

If $P_{bnb} = P_a = P_b = 100$

$$\Rightarrow D = 10000 - 2.5(100)^2 + 3(100) + 1.5(100)$$

$D = -14550$ dollars $\rightarrow$ Demand

Negative demand shows that the product prices are too high for the customers.

Task 2 & 3:

Task 2

We need to calculate the partial derivative wst P_{BnB} at $(100, 10, 10)$ to see the effect of P_{BnB} increase on demand.

$$D = 10000 - 2.5P_{BnB}^2 + 3P_A + 1.5P_B$$

$$\frac{\partial D}{\partial P_{BnB}} = 0 - 5P_{BnB} + 0 + 0$$

$$\left. \frac{\partial D}{\partial P_{BnB}} \right|_{(100, 10, 10)} = -5 \times 100 = -500$$

This shows the demand is very less, almost no demand.

Task 3

Now we need to calculate the partial derivative wst P_A at $(10, 100, 10)$ to see the effect of P_A increase on demand of P_{BnB} .

$$D = 10000 - 2.5P_{BnB}^2 + 3P_A + 1.5P_B$$

$$\frac{\partial D}{\partial P_A} = 0 - 0 + 3 + 0$$

$$\left. \frac{\partial D}{\partial P_A} \right|_{(10, 100, 10)} = 3$$

The demand of BnB would be ^{affected} 3 if the price of A & BnB kept constant and A is increased.

Task 4:

Task - 4:

$$D = 10000 - 2.5(P_{BnB})^2 + 3P_A + 1.5P_B \quad \text{--- (1)}$$

$$\frac{\partial D}{\partial P_{BnB}} = -5P_{BnB} \quad \text{--- (2)}$$

$$\Rightarrow \frac{\partial^2 D}{\partial P_{BnB}^2} = -5 < 0 \rightarrow \text{Shows that } D \text{ is maxima at critical point } P_{BnB}$$

② For critical point,

$$-5P_{BnB} = 0$$

$$P_{BnB} = 0$$

$$\text{①} \Rightarrow \frac{\partial D}{\partial P_A} = 3 \quad \left. \begin{array}{l} \text{---} \\ \text{---} \end{array} \right\} \text{So, no critical points.}$$

$$\text{③} \Rightarrow \frac{\partial D}{\partial P_B} = 1.5$$

Therefore, according to data,

D (demand) is maxima when the price, $P_{BnB} = 0$.

So, this shows that when price of $P_{BnB} = 0$, the demand is maximum. This is illogical but generally, the analysis shows inverse relation of demand and price of BnB product.

On the other hand, increase in prices of ^{product} A & B will increase the demand of BnB. However, if the prices decrease, the demand of BnB should also decrease in order to keep demand constant and cancel the effect of inverse and direct relation with demand. So, to increase demand of BnB, decrease price of BnB and increase price of A & B.

Code Flow Diagram

Importing Library



Printing Member's Name



User's consent to continue



Assuming the Prices of the Product



Calling Demand Automatic Differentiation function



Backend Automatic Computation of function concerning a variable



Returns result



Prints Result



Asking to run another time



If Yes, Continue...else, the Program terminates

Explanation of Major Software Commands Used

We've used Python as our platform of choice for writing the software. The main library that's been used is TensorFlow. This library was chosen due to its features of allowing the user to implement Automatic Differentiation functions like `GradientTape()/tape.watch()`. After importing the library, one of the major software commands is the declaration of a function **`Derivative(P_BnB, P_A, P_B)`**, taking in three float arguments which are **`P_BnB`**, **`P_A`**, and **`P_B`**.

Following this, another major software command is **`tf.GradientTape()`** where TensorFlow's `GradientTape()` function is called. This function produces a context for the implementation of Automatic Differentiation,

where operations involved with Automatic Differentiation can be monitored. Another major software command is **tape.watch()** which instructs the library to monitor the previous tensor argument which is doing the calculations in the backend.

Another very important software command that has been used is the **tape.gradient()** function whose parameters include a function based on the demand and **P_BnB**. This function computes the gradient. Lastly, another major software command that's been used is the usage of the original Derivative function which calculates the effect of the price changes on the respective products.

Example to run the Program

There are two major code-based parts of the Project. One deals with Task 2 and the other deals with Task 3, where both Tasks analyze the effect of an increase in BnB's price on its demand and analyze the effect of an increase in Product A's price on the demand of BnB respectively.

One needs to download TensorFlow for the usage of this specific program. The program can be run on any major code editor with the use of Python which can be downloaded from the Microsoft office or as an APK from Python's own website. When you're done with these prerequisites, you can run the program easily.

```
Welcome To The FMCG Company!  
M. Abdullah Siddiqui 23I-0617  
Moiz Ansari          23I-0523  
Faizan Ur Rehman     23I-0321  
Press Any Key To Continue.....
```

On the console, after the introduction of the group members, you'll be prompted to "Press any key to continue...", to which the user can press any key for the program to start running. After this, the program will run and ask the user to input price of BnB, and then use Automatic Differentiation to calculate how much the demand for BnB is affected by an increase in the price of the Product.

Then the program will prompt you with a choice to run another query. If you choose to do so, you may input 'y', and if you want to terminate the program, input 'n'.

```
Enter the price of BnB: 100  
Due to increase in Price of BnB, Demand of BnB is Affected by : -500.0  
Do you want to run another query? (y/n): y  
Enter the price of BnB: 150  
Due to increase in Price of BnB, Demand of BnB is Affected by : -750.0  
Do you want to run another query? (y/n): n  
Terminating the program...
```

This is the due process for the Python script of Task 2.

The initial prerequisites remain the same for the Task 3 program, where the console output is the same except that the program outputs the effect on the demand of BnB due to an increase in the price of product A.

```
Enter the price of A: 20  
Due to increase in Price of A, Demand of BnB is Affected by : 3.6  
Do you want to run another query? (y/n): y  
Enter the price of A: 30  
Due to increase in Price of A, Demand of BnB is Affected by : 3.6  
Do you want to run another query? (y/n): n  
Terminating the program...
```

Conclusions and Analysis

In the end, the problem is multifaceted where the demand for one product, BnB, is not only affected by the increase or decrease of its price, but also the changes in the prices of its competitors, which are Product A and Product B. The final conclusion is that all these factors do affect the demand for BnB significantly, with an increase in the price of BnB lowering the demand by -600, whereas an increase in the price of Product A, increasing the demand for BnB by 3.6.

Thus, the final suggestion for the company is to take measures that increase the price of product A/B and decrease the price of BnB itself. This could be done through more effective marketing strategies, and outsourcing various parts of the process of BnB manufacturing to lower the costs of its production and thus, increase its demand exponentially.

Contributions

Abdullah Siddiqui: Task 2 and Task 3 which include writing the code and doing the by-hand solutions.

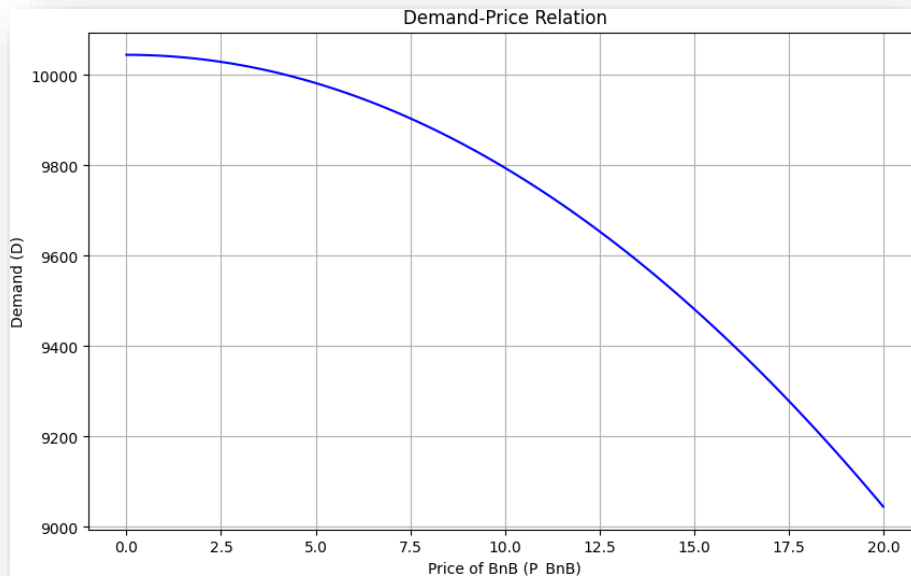
Moiz Ansari: Task 1 and Task 4 which includes forming the equation and giving suggestions to the company.

Faizan Ur Rehman Khan: Forming the entirety of the report, including methodology, detailing all solutions.

Difficulties faced

One of the major problems faced was the formation of the demand equation itself, which was followed by researching Automatic Differentiation and all the methods it could implement. This was resolved through extensive trial and error. Looking through different Python libraries in order to choose which one was the most ideal for our usage was also a difficult task but it was overcome with a joint research effort.

2D-Plot



For prices, $P_A = P_B = 10$

Annexure A

Task 2:

```
import tensorflow as tf # Importing TensorFlow Library for Automatic Differentiation

print("Welcome To The FMCG Company!")
print("M. Abdullah Siddiqui 23I-0617")
print("Moiz Ansari 23I-0523")
print("Faizan Ur Rehman 23I-0321")
input("Press Any Key To Continue.....") # Generate response according to user Input

def Derivative(P_BnB, P_A, P_B): # User Defined Function For Calculating Derivative
    with tf.GradientTape() as tape: # Used for Automatic Differentiation Based on
        inputs
            tape.watch(P_BnB) # Track operations involving P_BnB
            Function = 10000 - 2.5 * tf.pow(P_BnB, 2) + 3 * P_A + 1.5 * P_B # Making
Function

    Change = tape.gradient(Function, P_BnB) # Used for Automatic Differentiation
Based on inputs

    return Change # Returning The result of the defined function to the Called Place

def main():
    while True:
        # Assumed Values For Task 2
        P_A = 10.0 # Same Price Of A & B Products
```

```

P_B = 10.0
P_BnB = float(input("Enter the price of BnB: ")) # Take input for Pbnb from
the user

# Calculating The Effect
Effect = Derivative(tf.constant(P_BnB, dtype=tf.float32), tf.constant(P_A,
dtype=tf.float32), tf.constant(P_B, dtype=tf.float32)) # Calling The Defined Function

print("Due to increase in Price of BnB, Demand of BnB is Affected by : ",
Effect.numpy()) # Printing The Results

user_input = input("Do you want to run another query? (y/n): ").lower()
if user_input == "n":
    print("Terminating the program...")
    break

if __name__ == "__main__":
    main()

```

Task 3:

```

import tensorflow as tf # Importing TensorFlow Library for Automatic Differentiation
print("Welcome To The FMCG Company!")
print("M. Abdullah Siddiqui 23I-0617")
print("Moiz Ansari 23I-0523")
print("Faizan Ur Rehman 23I-0321")
input("Press Any Key To Continue.....") # Generate response according to user Input

def Derivative(P_BnB, P_A, P_B): # User Defined Function For Calculating Derivative
    with tf.GradientTape() as tape: # Used for Automatic Differentiation Based on
inputs
        tape.watch(P_A) # Track operations involving P_BnB
        Function = 10000 - 3 * tf.pow(P_BnB, 2) + 3.6 * P_A + 1.8 * P_B # Making
Function

        Change = tape.gradient(Function, P_A) # Used for Automatic Differentiation Based
on inputs

        return Change # Returning The result of the defined function to the Called Place

def main():
    while True:
        # Assumed Values For Task 2
        # Different Price Of A Product
        P_A = float(input("Enter the price of A: ")) # Take input for Pbnb from the
user
        P_B = 10.0 # Same Price of BnB and B
        P_BnB = 10.0

        # Calculating The Effect
        Effect = Derivative(tf.constant(P_BnB, dtype=tf.float32), tf.constant(P_A,
dtype=tf.float32), tf.constant(P_B, dtype=tf.float32)) # Calling The Defined Function

        print("Due to increase in Price of A, Demand of BnB is Affected by : ",
Effect.numpy()) # Printing The Results

        user_input = input("Do you want to run another query? (y/n): ").lower()
        if user_input == "n":
            print("Terminating the program...")

```

```
break
```

```
if __name__ == "__main__":  
    main()
```